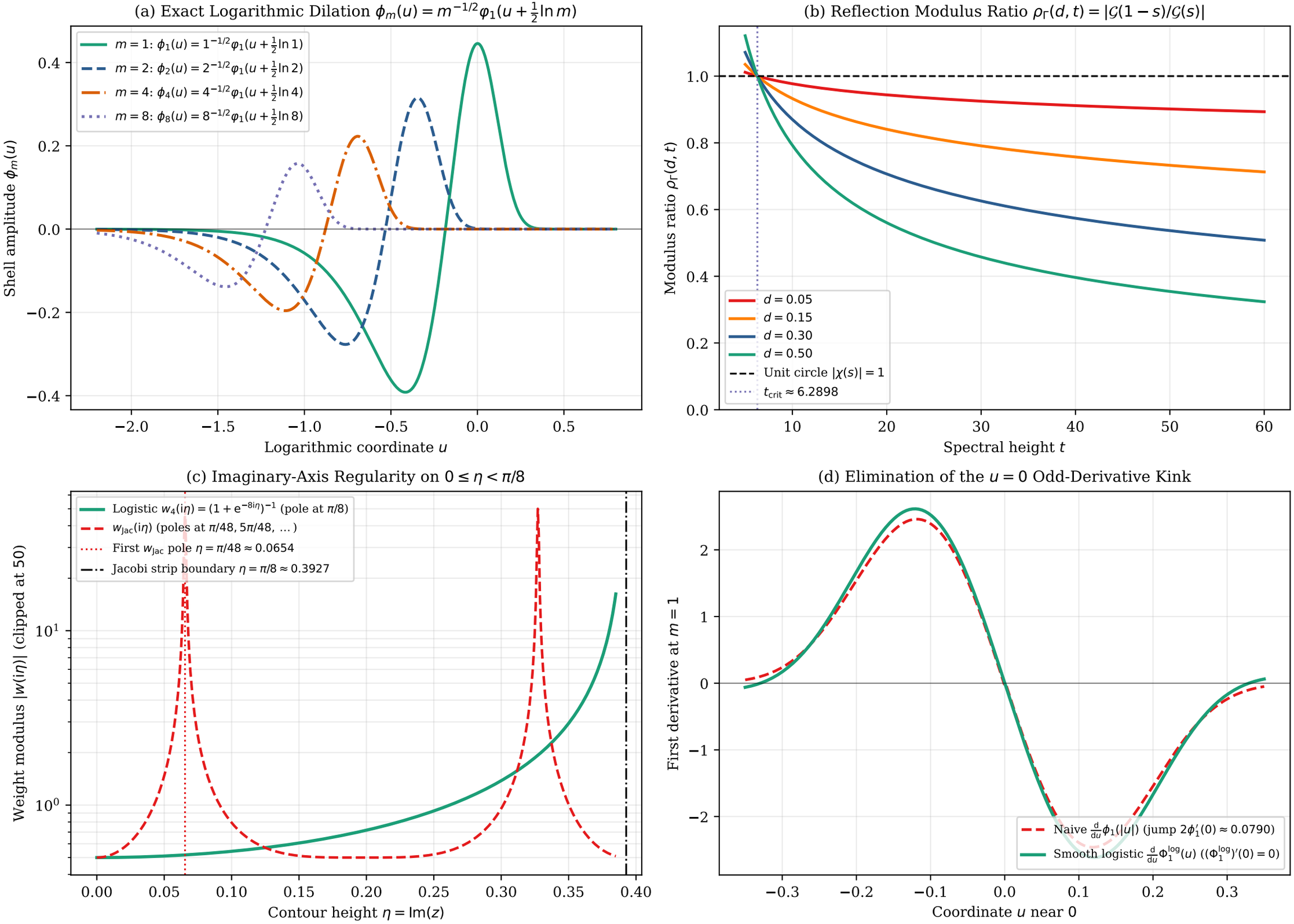

(a) Exact Logarithmic Dilation $\phi_m(u) = m^{-1/2}\varphi_1(u + \frac{1}{2}\ln m)$
$m = 1$: $\phi_1(u) = 1^{-1/2}\varphi_1(u + \frac{1}{2}\ln 1)$
$m = 2$: $\phi_2(u) = 2^{-1/2}\varphi_1(u + \frac{1}{2}\ln 2)$
$m = 4$: $\phi_4(u) = 4^{-1/2}\varphi_1(u + \frac{1}{2}\ln 4)$
$m = 8$: $\phi_8(u) = 8^{-1/2}\varphi_1(u + \frac{1}{2}\ln 8)$
Shell amplitude $\phi_m(u)$
Logarithmic coordinate u
(b) Reflection Modulus Ratio $\rho_\Gamma(d, t) = |\mathcal{G}(1 - s)/\mathcal{G}(s)|$
$d = 0.05$
$d = 0.15$
$d = 0.30$
$d = 0.50$
Unit circle $|\chi(s)| = 1$
$t_{\text{crit}} \approx 6.2898$
Modulus ratio $\rho_\Gamma(d, t)$
Spectral height t
(c) Imaginary-Axis Regularity on $0 \le \eta < \pi/8$
Logistic $w_4(i\eta) = (1 + e^{-8i\eta})^{-1}$ (pole at $\pi/8$)
$w_{\text{Jac}}(i\eta)$ (poles at $\pi/48, 5\pi/48, \ldots$)
First w_{Jac} pole $\eta = \pi/48 \approx 0.0654$
Jacobi strip boundary $\eta = \pi/8 \approx 0.3927$
Weight modulus $|w(i\eta)|$ (clipped at 50)
Contour height $\eta = \text{Im}(z)$
(d) Elimination of the $u = 0$ Odd-Derivative Kink
Naive $\frac{d}{du}\phi_1(|u|)$ (jump $2\phi_1'(0) \approx 0.0790$)
Smooth logistic $\frac{d}{du}\Phi_1^{\log}(u)$ ($(\Phi_1^{\log})'(0) = 0$)
First derivative at $m = 1$
Coordinate u near 0